10 In 2016 the population of the UK was 6.5×10^7

Laura wants to calculate an estimate for the population of the UK in 2020
She assumes that the population increases by 0.6% each year.

(a) Using Laura's assumption, calculate an estimate for the population of the UK in 2020

2016 to2020:four yearly increases.

$65000000 \times 1.006^4 = 66574096.24424$

$\approx 6.66 \times 10^7$ people (3 significant figures).

1 mark
Compound multiplier

1 mark
About66.6million

(2)

Kieran also assumes that the population of the UK increases by 0.6% each year.

He says that it will take over 80 years for the population to increase by 50% because $\frac{50}{0.6} = 83.\dot{3}$

Kieran's method is wrong.

(b) Explain what is wrong with his method.

Growth compounds:0.6% acts on a larger population each year.

He treats the yearly numerical increase as constant.

1 mark
Changing yearly base

(1)

Assuming that the population of the UK increases by 0.6% each year,

(c) show that the population of the UK each year forms a geometric progression.

New population = Old + 0.006×Old.

1 mark
Common ratio1.006

New population = 1.006×Old.

Every consecutive ratio is1.006, so it is geometric.

1 mark
Same ratio every year

(2)

(Total for Question 10 is 5 marks)